ML Training Dataset

Feature Reference Document

ml_training_dataset_extended.csv — 66 Columns — Sprint 11, Month 4

Sri Lanka Travel Recommendation System

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1. Overview

This document provides the complete data lineage for every column in ml_training_dataset_extended.csv. For each column it records: the source file it originates from, whether it is a direct copy or a calculation, and the exact formula used to derive it.

Property	Value	Notes
Total columns	66	8 meta + 105 ML features + 1 label
Total rows	3,000	29-day event window (Feb 25 – Mar 27, 2026)
Null values	0	All nulls filled per column logic below
Source: synthetic_dataset	Cols 1–8, 9–49, 60-64	User interactions & events
Source: destination_ref	Cols 50–59	Real Sri Lankan destination data
Label distribution	74% positive / 26% negative	Moderate 3:1 imbalance

2. Source Files

2.1 ml_training_dataset_extended.csv

The primary interaction log. Contains one row per user-destination event across 200 unique users and 100 destinations. Original columns used as base features or as inputs to aggregation calculations.

2.2 srilanka_destination_ref.csv

The destination item feature file. Contains one row per destination (100 rows) with real Sri Lankan destination metadata, multi-category scores, and engagement metrics derived from the interaction log. Joined to the training dataset on item_id = dest_id.

3. Event Weight Encoding

Used in: event_weight (col 6), recency_weight (col 7), interest_[cat] calculations (cols 9–15), interaction_weight_sum (col 109). Also the recommended sample_weight parameter during model training.

event_type	event_weight	Rationale
book	4	Strongest positive intent — financial commitment
save_trip	3	Strong intent — added to planned itinerary
save	2	Moderate intent — bookmarked for later
trip_click	1	Mild interest — destination card opened
view	0.5	Weak signal — passive impression

4. Full Column Reference

Colour coding: Blue = meta/temporal | Purple = destination item features | Green (dark) = G4 trip category usage | White/light = other feature groups.

Meta & Temporal Columns (1–8)

#	Column Name	Type / Range	Calculation / Formula
1	user_id	String ID	unique user identifier.
2	item_id	String ID	unique destination identifier. Joins to srilanka_destination_ref.dest_id.
3	event_type	Categorical	one of: view, trip_click, save, save_trip, book.
4	created_at	Timestamp (UTC)	ISO 8601 timestamp of the event.
5	destination_category	Categorical	primary category of the destination (History, Nature, Culture, Beach, Sightseeing, Adventure, Relaxation).
6	event_weight	Float {0.5,1,2,3,4}	Calculated: book=4 | save_trip=3 | save=2 | trip_click=1 | view=0.5. Encodes engagement strength. Use as sample_weight in training — NOT as an input feature.
7	recency_weight	Float 0.22–1.0	Calculated: exp(−0.05 × days_since_event). Events from 0 days ago = 1.0, 29 days ago ≈ 0.22. Use as sample_weight multiplied with event_weight.
8	days_ago	Float 0–30	Calculated: (max(created_at) − created_at) in decimal days. Used to derive recency weight.

G1 — User Interest Profile (9–16)

Answers: What does this user care about? How strongly? How broadly?

#	Column Name	Type / Range	Calculation / Formula
9	interest_sightseeing	Float 0.0–1.0	Calculated (per user, aggregated): Step 1 — weighted mean of Sightseeing_dest_score across all user events, weight = event_weight. Step 2 — Min-Max scaled across all 200 users so 0=lowest relative sightseeing interest, 1=highest.
10	interest_culture	Float 0.0–1.0	Same as interest_sightseeing — weighted mean of Culture_dest_score × event_weight, then Min-Max scaled across users.

#	Column Name	Type / Range	Calculation / Formula
11	interest_history	Float 0.0–1.0	Same — weighted mean of History_dest_score × event_weight, Min-Max scaled.
12	interest_nature	Float 0.0–1.0	Same — weighted mean of Nature_dest_score × event_weight, Min-Max scaled.
13	interest_beach	Float 0.0–1.0	Same — weighted mean of Beach_dest_score × event_weight, Min-Max scaled.
14	interest_adventure	Float 0.0–1.0	Same — weighted mean of Adventure_dest_score × event_weight, Min-Max scaled.
15	interest_relaxation	Float 0.0–1.0	Same — weighted mean of Relaxation_dest_score × event_weight, Min-Max scaled.
16	interest_strength	Float 0.08–0.39	Calculated: max(raw_pref_vector) − mean(raw_pref_vector). High value = specialist user strongly focused on one category. Low value = generalist with broad, balanced interests.

G2 — Preference Patterns (17–24)

Answers: How does this user behave? Are they a browser or a booker? Loyal or explorative?

#	Column Name	Type / Range	Calculation / Formula
17	booking_rate	Float 0.0–0.5	Calculated (per user): COUNT(event_type='book') / COUNT(all events). Proportion of interactions that are direct bookings. High = decisive, intent-driven user.
18	save_rate	Float 0.0–0.6	Calculated: COUNT(save + save_trip) / COUNT(all). Proportion of interactions that are saves. High = shortlister who curates before committing.
19	exploration_rate	Float 0.0–0.9	Calculated: COUNT(view) / COUNT(all). Proportion that are passive views. High = browser, low = intent-driven visitor.
20	conversion_rate	Float 0.0–0.6	Calculated: COUNT(book + save_trip) / COUNT(all). Overall strong-commitment rate. Best combined signal of user decisiveness.
21	travel_pace_preference	Ordinal 0/1/2	Calculated: mean(event_weight) per user bucketed — RELAXED=0 (mean ≤1.5, view-heavy), MODERATE=1 (1.5–2.5), PACKED=2 (>2.5, book-heavy). No users reached PACKED in this 29-day window.
22	category_loyalty_score	Float 0.19–0.57	Calculated: max(category_interaction_share) per user. Highest share any single category takes of a user's total interactions. High = loyal to one type, low = diverse explorer.
23	user_unique_dests	Integer 6–26	Calculated: COUNT(DISTINCT item_id) per user. Number of distinct destinations interacted with.
24	user_active_days	Integer 1–29	Calculated: (MAX(created_at) − MIN(created_at)).days + 1 per user. Span of engagement across the window.

G3 — Feedback Ratings (25–35)

Answers: How positive is this user overall? Which categories do they rate well?

#	Column Name	Type / Range	Calculation / Formula
25	feedback_positivity_rate	Float 0.0–1.0	Calculated (per user): COUNT(has_positive_feedback=1) / COUNT(all events). Global fallback = 0.4157 for users with no feedback. Corr with label: 0.295.
26	fb_rate_sightseeing	Float 0.0–1.0	Calculated: positive feedback rate filtered to events where destination_category='Sightseeing'. Fallback to global mean (0.4157) if user has no sightseeing interactions.
27	fb_rate_culture	Float 0.0–1.0	Same — feedback positivity rate for Culture category interactions. Corr with label: 0.218.
28	fb_rate_history	Float 0.0–1.0	Same — History category. Corr with label: 0.205.
29	fb_rate_nature	Float 0.0–1.0	Same — Nature category. Strongest per-category signal: corr 0.228.
30	fb_rate_beach	Float 0.0–1.0	Same — Beach category. Corr: 0.211.
31	fb_rate_adventure	Float 0.0–1.0	Same — Adventure category. Corr: 0.152.
32	fb_rate_relaxation	Float 0.0–1.0	Same — Relaxation category. Note: very few interactions (26 rows total), so fallback triggers often.
33	dest_feedback_rate	Float 0.12–0.71	Calculated (per destination): COUNT(has_positive_feedback=1) / COUNT(all events for this dest). Item-side signal. Corr: 0.113.
34	dest_pos_feedback_count	Integer	Calculated: COUNT(has_positive_feedback=1) grouped by item_id.
35	dest_total_feedback	Integer	Calculated: COUNT(all events) grouped by item_id. Denominator for dest_feedback_rate.

G4 — Trip Category Usage (36–49)

Answers: Which categories does this user actually book vs just browse? How recently? How efficiently do they convert views into bookings per category?

#	Column Name	Type / Range	Calculation / Formula
36	cat_usage_share_sightseeing	Float 0.0–1.0	Calculated: COUNT(all event types where dest_category='Sightseeing') / COUNT(all events) per user.
37	cat_usage_share_culture	Float 0.0–1.0	Same formula for Culture.
38	cat_usage_share_history	Float 0.0–1.0	Same formula for History.
39	cat_usage_share_nature	Float 0.0–1.0	Same formula for Nature.
40	cat_usage_share_beach	Float 0.0–1.0	Same formula for Beach.
41	cat_usage_share_adventure	Float 0.0–1.0	Same formula for Adventure.
42	cat_usage_share_relaxation	Float 0.0–1.0	Same formula for Relaxation.
43	cat_recency_sightseeing	Float 0.0–1.0	Calculated: MIN(days_ago) for user events where dest_category='Sightseeing', normalised by 29 (window length). 0=interacted today, 1=29 days ago or never.
44	cat_recency_culture	Float 0.0–1.0	Same recency formula for Culture.

#	Column Name	Type / Range	Calculation / Formula
45	cat_recency_history	Float 0.0–1.0	Same for History.
46	cat_recency_nature	Float 0.0–1.0	Same for Nature.
47	cat_recency_beach	Float 0.0–1.0	Same for Beach.
48	cat_recency_adventure	Float 0.0–1.0	Same for Adventure. Mean=0.50 (moderate recency).
49	cat_recency_relaxation	Float 0.0–1.0	Same for Relaxation. Mean=0.96 — most users rarely or never touched Relaxation category.

Destination Item Features (50–59)

All sourced from srilanka_destination_ref.csv joined on item_id = dest_id.

#	Column Name	Type / Range	Calculation / Formula
50	aggregate_feedback_rating	Float 4.02–4.85	Direct copy of bayesian_avg_rating from destination_ref..Pre-computed Bayesian average rating that penalises destinations with fewer than 5 reviews. Grounded in real Sri Lankan destination prominence.
51	booked_users_norm	Float 0.1–1.0	Calculated: booked_users / MAX(booked_users). Unique users who booked this destination, normalised.
52	weighted_engagement_norm	Float 0.30–1.0	Calculated: weighted_engagement / MAX(weighted_engagement). Sum of (book×4 + save_trip×3 + save×2 + trip_click×1 + view×0.5) per destination, normalised.
53	Sightseeing_dest_score	Float 0.0–0.90	Multi-category relevance score. Reflects how much sightseeing value this destination offers. Non-zero for 94.4% of rows.
54	Culture_dest_score	Float 0.0–0.95	Culture relevance. Non-zero for 47.4% of rows.
55	History_dest_score	Float 0.0–0.96	History relevance. Non-zero for 45.5% of rows.
56	Nature dest score	Float 0.0–0.92	Nature relevance. Non-zero for 61.8% of rows.
57	Beach_dest_score	Float 0.0–0.89	Beach relevance. Non-zero for 22.7% of rows.
58	Adventure_dest_score	Float 0.0–0.87	Adventure relevance. Non-zero for 42.6% of rows.
59	Relaxation_dest_score	Float 0.0–0.84	Relaxation relevance. Non-zero for 33.7% of rows.

Interaction & Derived Features (60–64)

Row-level and user-destination pair features capturing click behaviour, selection rate, and personalisation alignment.

#	Column Name	Type / Range	Calculation / Formula
60	click_frequency	Integer 0-2	Calculated (per user-destination pair): COUNT(event_type='trip_click'). 0 if no clicks. 97% of pairs have 0 or 1 click. Corr with label: 0.221.
61	user_item_affinity	Float 0.10–2.61	Calculated: dot product of user's interest vector × destination's category score vector. Formula: SUM(interest [cat] × [Cat] dest score) for all 7

#	Column Name	Type / Range	Calculation / Formula
			categories. Scalar alignment between user taste and destination character. Mean=1.19, corr=0.076 (non-linear interactions expected to be higher in tree models).
62	category_diversity	Float 0.0–1.0	row-level diversity signal from source data.
63	trust_score	Float 0.30–0.89	row-level user trust score from source data.
64	user total feedback count	Integer 1–50	total lifetime feedback count for the user.

Label (65)

#	Column Name	Type / Range	Calculation / Formula
65	label	Binary 0/1	1 = positive engagement (book, save, save_trip, trip_click). 0 = negative/passive (view events with no downstream engagement). Distribution: 2221 positive (74%), 779 negative (26%).

5. Key Calculation Formulas

5.1 interest_[cat] (cols 9–15)

```
Step 1 — Raw weighted preference per user:
  raw_pref_[cat] = SUM([Cat]_score × event_weight) / SUM(event_weight)
  grouped by user_id across all events

Step 2 — Min-Max scaling across all users:
  interest_[cat] = (raw_pref_[cat] - MIN_users) / (MAX_users - MIN_users)
```

Example (user_182, dest_38 row): raw_pref_Sightseeing = (Σ Sightseeing_score × event_weight) / Σ event_weight = 0.5878 → scaled to 0.7476

5.2 feedback_positivity_rate (col 25)

```
  feedback_positivity_rate = COUNT(has_positive_feedback=1) / COUNT(all rows)
  grouped by user_id
  fallback = 0.4157 (global mean) if user has no history
```

5.3 cat_recency_[cat] (cols 43–49)

```
  min_days_ago = MIN(days_ago) for user events where dest_category=[cat]
  cat_recency_[cat] = min_days_ago / 29.0   (normalised to window length)
  0 = interacted today, 1 = 29 days ago or never touched
```

5.4 user_item_affinity (col 111)

```
  user_item_affinity = DOT(interest_vector, score_vector)
  = Σ interest_[cat] × [Cat]_score   for cat in all 7 categories
```

Example (user_182, dest_38): interest_vector=[0.748,0,0,0.462,0.523,0.352,0.648] × score_vector=[0.70,0,0,0.58,0,0.35,0.75] = 1.4095

5.5 recency_weight (col 7)

```
recency_weight = exp(-0.05 × days_since_event)
days_since_event = (MAX(created_at) - created_at) in decimal days
Range: 1.0 (today) → 0.22 (29 days ago)
Recommended use: sample_weight = event_weight × recency_weight
```

6. Notes

- event_weight and recency_weight are NOT input features — pass as sample_weight to avoid data leakage (they directly determine the label).
- Class imbalance 74/26: use class_weight='balanced' in sklearn or scale_pos_weight in XGBoost. Evaluate with F1 and AUC-PR.
- Relaxation is critically underrepresented (1 destination, 26 rows). Handle via hybrid rules layer in production rather than dataset augmentation.
- Recommended split: time-based — train Feb 25–Mar 17 (~70%), val Mar 18–22 (~15%), test Mar 23–27 (~15%). Stratify on label within splits.